EVIDENCE OF A MICROPLATE IN THE SOUTHERN ANDES ?

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ABSTRACT

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Paleomagnetic poles for Late Paleozoic-Early Mesozoic rocks exposed in stable and possible mobile zones of Western Argentina are reported. They suggest two interpretations.

One of the interpretations suggests that a mobile zone situated at about 32.1°South 69.3°West, in the Central Andes, rotated about 60° clockwise after Late Paleozoic time.

The other interpretation suggests that at least the Andean zone between 32.1° and 31.7° South is allochthonous and was situated in the Pacific, at tropical latitudes, in the Late Paleozoic. On this interpretation the accretion of this microplate to South America ocurred in Permo-Triassic times.

INTRODUCTION

Late Paleozoic and Early Mesozoic rocks are exposed in both stable and unstable (or possible mobile) zones of South America. Until now paleomagnetic studies have been carried out in Late Paleozoic-Early Mesozoic rocks from stable zones of South America (Valencio and Villas, 1972; Valencio and Mitchell, 1972; Valencio, 1973; Vilas, 1976; Vilas and Valencio, 1978 and Vilas, 1980). In this paper we report paleomagnetic poles for Late Paleozoic-Early Mesozoic rocks from two new stables zones and two possible mobile zones from Argentina. We also discuss interpretations suggested by these paleomagnetic poles concerning possible geodynamics of the mobile zones.

THE PALEOMAGNETIC AND RADIOMETRIC DATA

Paleomagnetic studies were carried out on four sequences of Late Paleozoic-Early Mesozoic sediments and igneous rocks from western Argentina. Thermal, alternating field and chemical treatment were used to

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define the stable remanent magnetization (s.r.m.) of the samples (Valencio and Vilas, in press). Structural corrections were used to obtain ancient geomagnetic field directions from samples from some of the sequences.

K/Ar^{40} age determinations were carried out on some of the igneous rocks to more precisely define the age of these rocks.

We were requested that this paper emphasize conclusions, models and speculations as opposed to detail data sets. Therefore, in the following sections we do not present a complete discussion of the laboratory techniques used or of the paleomagnetic and radiometric data we obtained. Further details concerning these will be published elsewhere (Valencio and Vilas, in press). However, some statistics of the reported paleomagnetic data are included (Table I) to help readers evaluate our interpretations.

Stable zones

Sampling was carried out in sequences of sediments (Guandacol Formation) and igneous rocks (Cochicó Formation) at two different stable zones.

Seventy nine samples were collected from a sequence of sediments of the Guandacol Formation, about 700 m thick, exposed in Sierras Pampeanas (Site 1, Fig. 1) at about 29.5°S 68.5°W. The age of these sediments is Early Carboniferous. Tectonic corrections were applied to the primary remanent magnetizations of these samples. The statistics of paleomagnetic data and the paleomagnetic pole (PP) for this formation (C_3) are given in Table I; the PP is shown in Fig. 2.

Cochicó Formation is the local name given to the igneous rocks included in the Choiyoilitense Group. Rocks of this group are exposed in many sites of the Andean and extra-Andean zone of Argentina. One hundred and twenty nine samples of the Cochicó Formation were collected in the southern extra-Andean zone of Province of Mendoza (Site 2, Fig. 1) at about 34.7°S 68.6°W. The age of the igneous rocks of Cochicó Formation is late Early Permian (Sakmarian-Artinskian; about 260 m.y.). The statistics of the paleomagnetic data and the PP for these rocks (P_6) are given in Table I. A fold test indicates that the origin of the s.r.m. of the Cochicó rocks is pretectonic. The PP for this formation is also shown in Fig. 2.

Mobile zones?

Sampling was carried out in sequences of sediments (Yalguaraz Formation) and igneous rocks of the Choiyoilitense Group at two different possible mobile zones.

A sequence of sediments of the Yalguaraz Formation is exposed in the

TABLE I

Paleomagnetic data for Late Paleozoic-Early Mesozoic rocks from Argentina reported in this paper (△, s.r.m. of normal polarity; ▲, s.r.m. of reversed polarity; ◭, s.r.m. of normal and reversed polarities).

Site	Sampled Site		Formation	Age	Stable Remanence referred to paleohorizontal						Paleomagnetic Pole					Symbol Name
	Lat (°S)	Long (°W)			N	D (°)	I (°)	k	R	α_{95} (°)	Lat (°S)	Long (°E)	k	R	A_{95} (°)	
4	31.7	69.5	Quebrada de la Alumbrera (Choiyoilitense Group)	Permo-Triassic	21	183	+70	20	19.990	7	67	290	11	19.213	10	◭ P_{12}
2	34.7	68.6	Cochicó (Choiyoilitense Group)	late Early Permian	43	12	−42	18	40.697	5	77	163	18	40.599	5	◐ P_6
3	32.1	69.3	Portezuelo del Cenizo (Choiyoilitense Group)	Permo-Carboniferous or Early Permian	9	246	+42	41	8.807	8	32	211	32	8.747	9	▲ P_{13}
3	32.1	69.3	Yalguaraz	Mid-Carboniferous	19	37	−40	25	18.284	7	56	195	23	18.211	7	△ C_4
1	29.5	68.5	Guandacol	Early Carboniferous	27	197	+47	12	24.850	8	75	210	11	24.642	9	◐ C_3

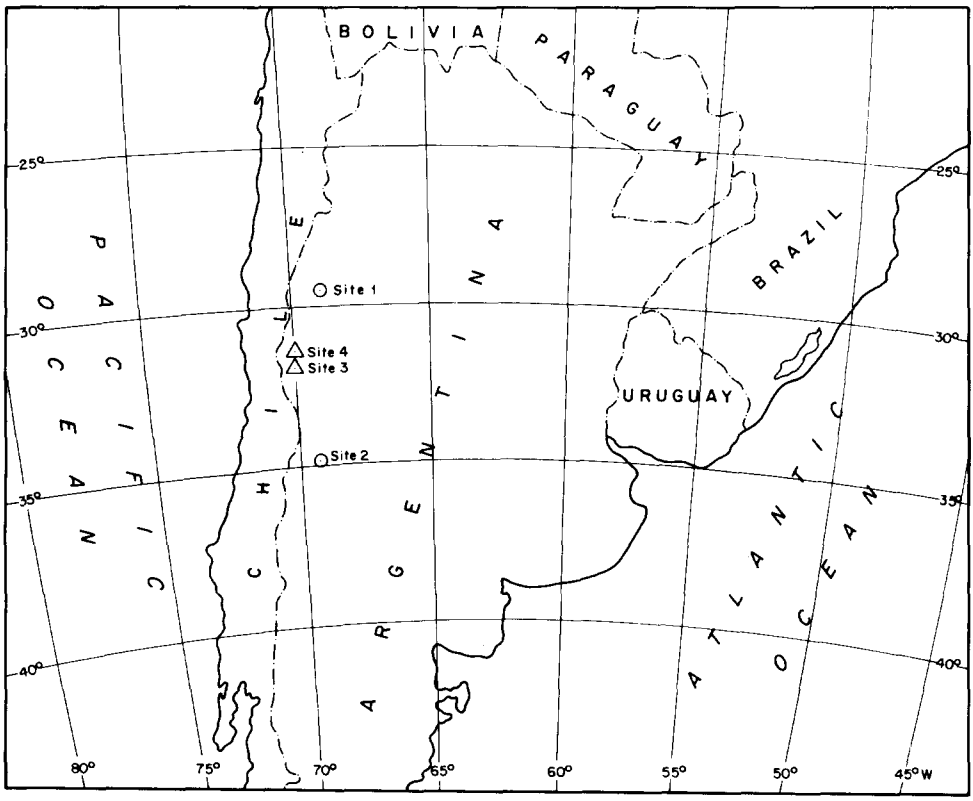


Fig. 1. Sampling sites. ○, site in a stable zone; △, site in a possible mobile zone.

Andean zone of the Province of Mendoza, Western Argentina, at about 32.1°S 69.3°W (Site 3, Fig. 1). A mid-Carboniferous(?) age was suggested for these sediments. The highest 30 m of the sequence of the Yalguaraz Formation was sampled (forty five hand samples). The statistics of the paleomagnetic data and the PP for this Formation (C_4) are given in Table I; the PP is shown in Fig. 2.

Igneous rocks of the Portezuelo del Cenizo Formation, included in the Choiyoilitense Group, are also exposed in Site 3 (Fig. 1). There the sequences of rocks of the Portezuelo del Cenizo Formation rests on an erosional unconformity which separates these igneous rocks from the underlying sediments of Yalguaraz Formation. A Permo-Carboniferous or Early Permian age has been suggested for these igneous rocks. The lowest 46 m of the sequence of Portezuelo del Cenizo Formation was sampled (twenty hand samples). The statistics of the paleomagnetic data and the PP for this Formation (P_{13}) are given in Table I; the PP is shown in Fig. 2.

Choiyoilitense igneous rocks are also exposed at the Quebrada de la

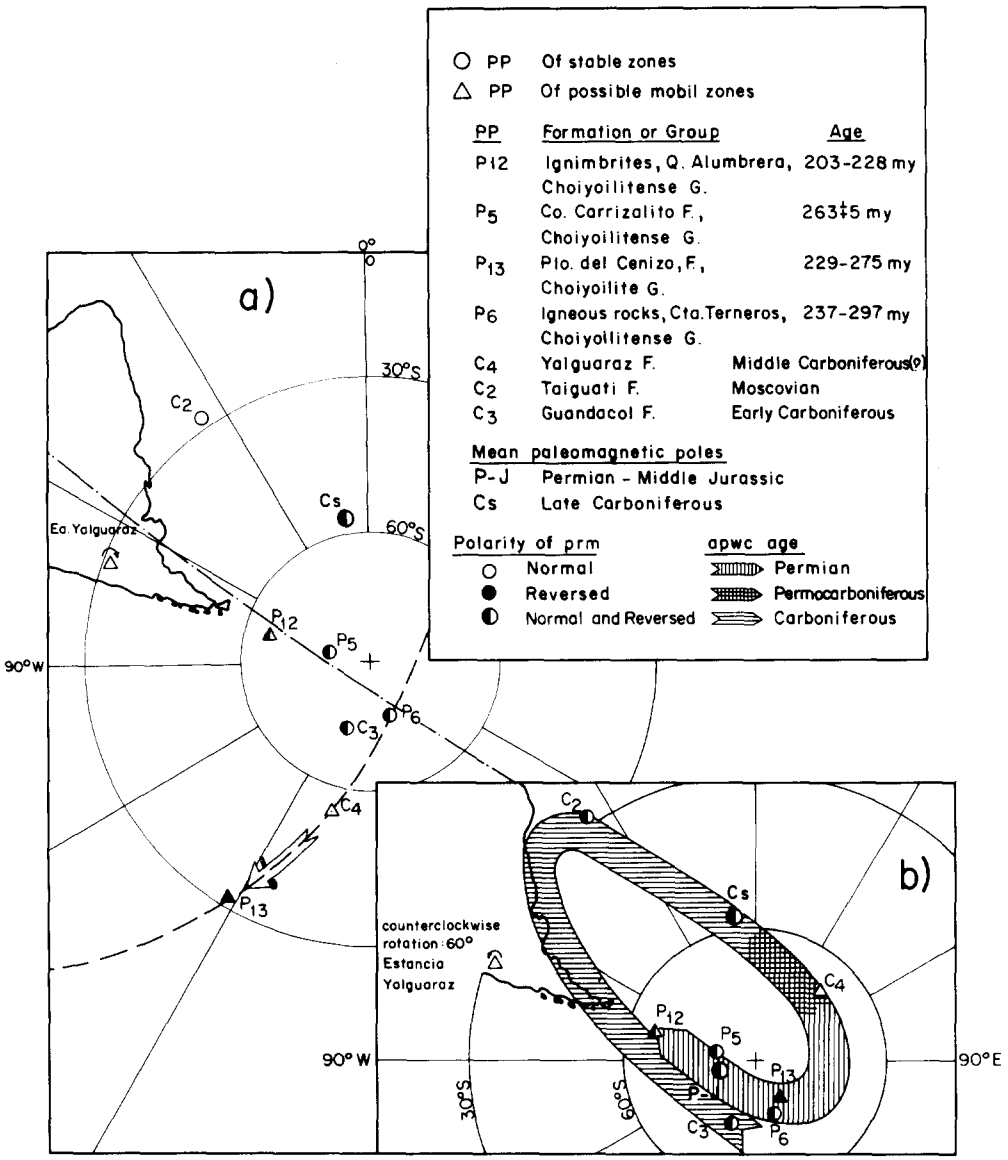


Fig. 2. Interpretation 1. Late Paleozoic paleomagnetic poles (PP) for South America. a) trends defined by PPs: – · –, great circle defined by PPs from Choiyoilitense Group rocks P_6 and P_{12} (Table I) and P_5 (Valencio and Mitchell, 1972), and, ---, small circle defined by PPs P_6, C_4 and P_{13} (Table I). b) Late Paleozoic apparent polar wandering curve for South America and relative positions of C_4 and P_{13} after a counterclockwise rotation of 60° around an axis situated in the Ea. Yalguaraz sampling site (Site 3, Fig. 1) (see text).

Alumbrera in the Andean zone of the Province of Mendoza, Western Argentina, at about 31.7°S 69.5°W (Site 4, Fig. 1). A Permo-Triassic age was suggested for these rocks. Sixty hand samples were collected covering a section of about 100 m of the sequence of the igneous rocks exposed in this site. The statistics of the paleomagnetic data and the PP for these igneous rocks (P_{12}) are given in Table I; the PP is shown in Fig. 2.

INTERPRETATION

The paleomagnetic data suggest two different interpretations. The only way to evaluate these two possibilities is to compare the consistency of the age of the PPs reported in this paper (Table I) with the "relative" age assigned to these PPs in the interpretations based on the Late Paleozoic-Early Mesozoic PPs for the Gondwanic continents. This explains why we emphasize in the "relative" ages assigned to the reported PPs in the following two interpretations.

Interpretation 1

Fig. 2a shows the positions of the PPs reported in this paper. In this figure some of the Late Paleozoic PPs from stable zones of South America are also shown. They are: C_2, Taiguati Formation, Middle Carboniferous; Cs, mean position of Late Carboniferous (Stephanian) PPs; and P_5, Cerro Carrizalito Formation (Choiyoilitense Group), Middle Permian (Valencio and Vilas, 1972; Valencio and Mitchell, 1972; Valencio, 1973; Vilas, 1976; Vilas and Valencio, 1978; Vilas, 1980).

Figure 2a also shows that the PPs for the Choiyoilitense rocks included in the Cochicó (P_6, Site 2) and Cerro Carrizalito Formations (P_5, Site 2) and exposed at Quebrada de la Alumbrera (P_{12}, Site 4) are aligned along a great circle. This great circle may be computed by least squares (Schomaker *et al.*, 1959). The pole of this circle is situated at 39°E 6°S ($\lambda = 0.002$). The value of λ indicates a very good fit of the PPs to this great circle (for an ideal, $\lambda = 0$; no fit, $\lambda = 1$). This great circle is 12° away from the sampling sites of those rocks. Figure 2a also shows that the PP for the Choiyoilitense rocks of the Portezuelo del Cenizo Formation (P_{13}, mobile zone?) is situated far away from that great circle. This suggests that the relative positions of P_6 (Site 2), P_5 (Site 2) and P_{12} (Site 4) are not caused by large composite tectonic movements of the sampling sites. We think that the Choiyoilitense igneous rocks used to determine these PPs were poured out in magmatic episodes of different age, during a period of fast polar wander relative to

South America in Late Permian- Permian Triassic time. These magmatic episodes would have ocurred in the following chronological order: Cochicó Formation (P_6, late Early Permian) → Cerro Carrizalito Group (P_5, late Early Permian) → Quebrada de la Alumbrera (P_{12}, Permo-Triassic).

The PP for the Choiyoilitense rocks of the Portezuelo del Cenizo Formation (P_{13}, Permo-Carboniferous or Early Permian) is displaced from the great circle defined by P_6-P_5-P_{12}. These rocks were collected in the same sampling site (Estancia Yalguaraz, Site 3) as those of the Yalguaraz Formation (C_4, Middle Carboniferous). Moreover, the sampling was carried out in the same sequences of rocks; an erosional uncomformity is between the sediments of the Yalguaraz Formation and the igneous rocks of the Portezuelo del Cenizo Formation. Thus, the positions of PPs C_4 and P_{13} (Fig. 2a) may define the Middle Carboniferous-Early Permian section of the apparent polar wandering curve for that sampling site (3). Fig. 2a shows that this section of the apparent polar wandering curve defined by these points and the PP P_6 (stable zone, Site 2) is a small circle centered at the sampling site 3 (this may also be defined computing the distance between P_6-Site 3 = 66.3°; P_{13}-Site 3 = 65.8°; and C_4-Site 3 = 66.8°; mean value 66.3° ± 0.4°). This suggests a clockwise rotation of the sampling site 3 post Early Permian time. That is, the pre-rotation positions of C_4 and P_{13} can be obtained by submitting them to a counterclockwise rotation around a vertical axis centered at the sampling site 3, until the best fit with the Late Paleozoic PPs from stable zones of South America is obtained. Fig. 2b shows these positions after a counterclockwise rotation of about 60°. They then fit well into the Carboniferous-Permian section of the apparent polar wandering curve of South America ($C_3 \rightarrow C_2 \rightarrow Cs \rightarrow C_4 \rightarrow P_{13}$,$P_6$) (Fig. 2b).

Fig. 3 shows the Late Paleozoic PPs and common section of the apparent polar wandering curve for South America, Africa, Australia, Madagascar and India. This section of the apparent polar wandering curve is consistent with that for South America (Fig. 2b). The positions of C_4 and P_{13} after a counterclockwise rotation of about 60° are also shown in Fig. 3. Figures 2b and 3 suggest that sampling site 3 (Estancia Yalguaraz) could have rotated 60° in a clockwise direction after Late Paleozoic time. However, we should note that this interpretation suggests either a Late Carboniferous or Permo-Carboniferous age for the Yalguaraz Formation (C_4), younger than the mid-Carboniferous (?) age suggested for this formation.

On the other hand the relative position of the PP for the Guandacol Formation (C_3, Early Carboniferous) shown in Fig. 3 suggests that this formation is older than the visean age of the Gilmore and Isismurra igneous rock from Australia (Cl_1, Fig. 3).

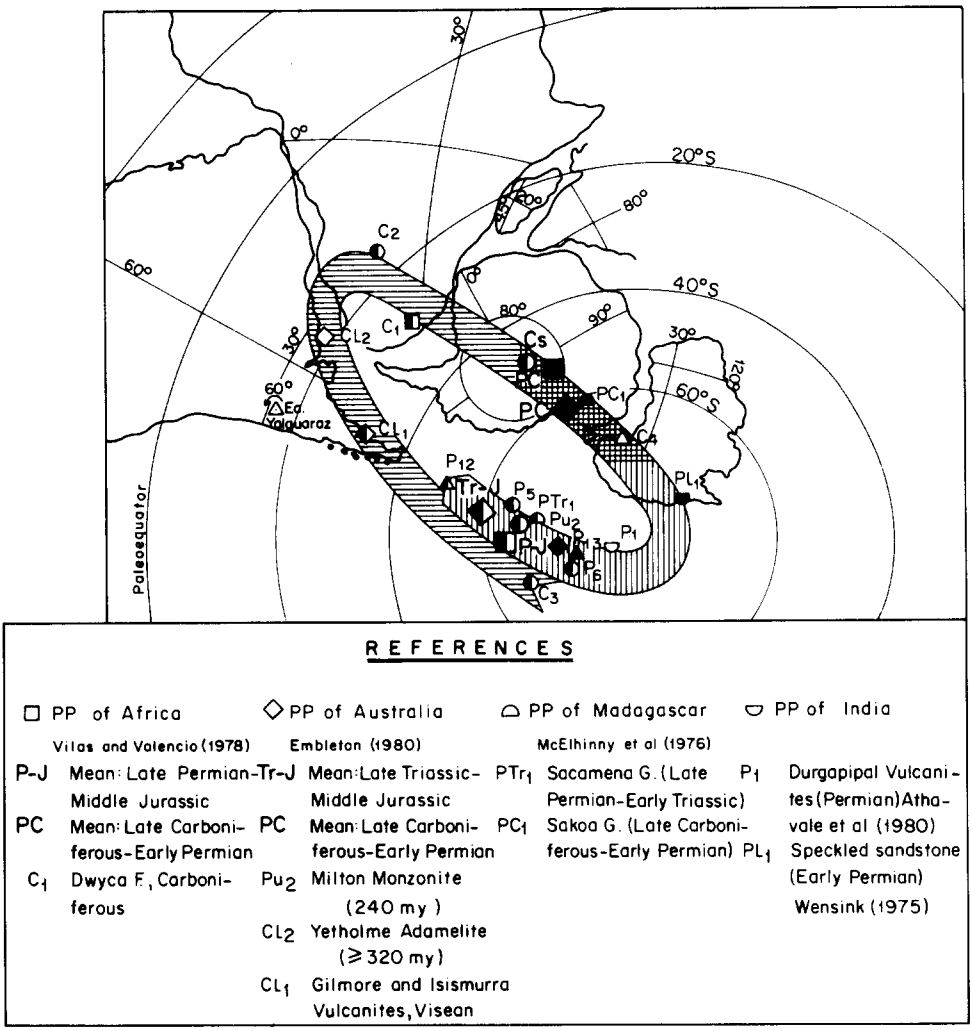


Fig. 3. Interpretation 1. Late Paleozoic and Early Mesozoic PPs and section of the apparent polar wandering curve for Gondwana. References in Fig. 2 and Table I. Reconstruction of Gondwana of Vilas and Valencio (1977).

Interpretation 2

In this interpretation we assume that sampling sites 3 and 4 were part of an allochtonous microplate in Late Paleozoic time. Therefore the PPs for the Late Paleozoic rocks exposed in these two sites (C_4 and P_{13} of the Yalguaraz and Portezuelo del Cenizo Formations, site 3, and P_{12} of the

Quebrada de la Alumbrera, site 4) should define an apparent polar wandering curve different from that defined by the Late Paleozoic PPs from the stable zones of the Gondwanic continents (Fig. 4a). To define the relative positions of the microplate and these stable zones in the Late Paleozoic we must overlap the PPs for the microplate with those of equivalent age for the Gondwanic continents. Fig. 4b shows the relative positions of the microplate and the Gondwana reconstruction in Late Paleozoic time; note that the PP for the Yalguaraz Formation (C_4, mid-Carboniferous ?) is close to the PPs for the Carboniferous Dwyka Formation (C_1, Africa) and the Moscovian Taiguati Formation (C_2, South America). It also shows that the PP for the Portezuelo del Cenizo Formation (P_{13}, Permo-Carboniferous or Early Permian) is close to the mean position of Permo-carboniferous PPs for Africa (PC) and Australia (PC) and the Permo-carboniferous PP for Madagascar (PC_1). The figure suggests that the microplate (dashed) was in the Pacific Ocean in Late Paleozoic time (perhaps forming part of the continent of Pacifica as proposed by Nur and Ben-Avraham, 1980, ?).

Fig. 4a shows that the accretion of the microplate to South America occured in the Permo-Triassic. This is suggested by the join of the apparent polar wandering curve of the microplate (P_{12}) with that of the Gondwana Supercontinent.

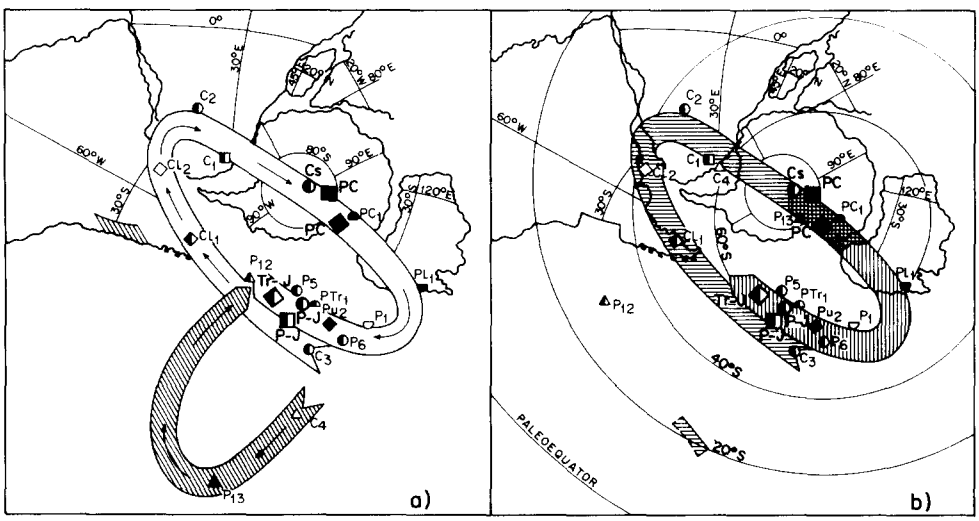


Fig. 4. Interpretation 2. Late Paleozoic and Early Mesozoic PPs for Gondwana (references Fig. 2 and 3, Table I). a) Dashed: possible accretional microplate at western South America and its Late Paleozoic apparent polar wandering curve. b) Relative position of the microplate in Late Paleozoic times obtained by overlapping its apparent polar wandering curve with that one of Gondwana.

DISCUSSION

Paleomagnetic data for Late Paleozoic-Early Mesozoic rocks exposed in stable and possible mobile zones of South America suggest two interpretations.

One of these interpretations suggests that a mobile zone situated in the Central Andes (32.1°S, 69.3°W) rotated 60° clockwise after Late Paleozoic time.

The other interpretation suggests that this zone is allochtonous and was situated in the Pacific, at tropical latitudes, in Late Paleozoic times. This latter interpretation is more consistent with the age of the rocks involved in this paleomagnetic study.

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